ModelIt!

Teacher's Guide

Why Your Hot Cocoa Betrays You

The Science of Thermal Energy Transfer

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-PS3-3, MS-PS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS3-3, MS-PS3-4

Apply scientific principles to design, construct, and test a device that either minimizes or maximizes thermal energy transfer. Plan an investigation to determine the relationships among the energy transferred, the type of matter, the mass, and the change in the average kinetic energy of the particles as measured by the temperature of the sample.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how temperature difference, insulation, surface area, and air movement affect thermal energy transfer rates.
Disciplinary Core Idea	**PS3.A Definitions of Energy / PS3.B Conservation of Energy and Energy Transfer** — Temperature is a measure of the average kinetic energy of particles. Energy transfers from hot to cold objects through conduction, convection, and radiation.
Crosscutting Concept	**Energy and Matter: Flows, Cycles, and Conservation** — Students trace the flow of thermal energy from the hot drink through the container to the surrounding environment, understanding that energy is conserved but transferred.

CAST Testing Alignment

MULTIPLE CHOICE:

A student places a metal spoon and a wooden spoon in the same cup of hot water. After 2 minutes, the metal spoon feels much hotter. Which best explains this observation?

CONSTRUCTED RESPONSE:

Design and describe a container that would keep an ice cube frozen for as long as possible. Explain how your design minimizes thermal energy transfer using conduction, convection, and radiation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Thermal energy transfer is one of the most fundamental and observable physics concepts. Every student has experienced a hot drink cooling down — this lesson connects that everyday experience to the science of energy flow.

Three Methods of Heat Transfer

Conduction: energy transfer through direct molecular contact (metal spoon in hot soup). Convection: energy transfer through fluid movement (warm air rising from a heater). Radiation: energy transfer through electromagnetic waves (feeling warmth from the sun). All three happen simultaneously with your hot cocoa.

Temperature and Particle Motion

Temperature measures the average kinetic energy of particles. In a hot drink, water molecules are moving fast. They transfer their kinetic energy to slower-moving air molecules through collisions. This is conduction at the molecular level. The drink cools because its particles slow down.

Insulation Science

Good insulators work by trapping pockets of air. Air is a poor conductor of heat, so trapped air layers create barriers to conduction. Foam cups, down jackets, and double-pane windows all use this principle. A vacuum (no air at all) is the best insulator — that's how a thermos works.

Real-World Applications

Thermal engineering is critical in building design (insulation, HVAC systems), spacecraft (extreme temperature management), electronics (heat sinks), and food safety (keeping hot foods hot and cold foods cold). Understanding heat transfer saves energy, money, and lives.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify temperature difference, insulation level, container surface area, and air movement as the key components affecting heat loss.

E — Establish Relationships
Students determine that insulation negatively affects heat loss (slows it down) while air movement and surface area positively affect heat loss (speed it up).

V — Visualize & Model
Students build a model showing how these four factors interact to determine how quickly a hot drink cools.

E — Evaluate Outcomes
Students run open mug vs. travel mug vs. windy day scenarios to quantify the impact of each variable.

R — Revise & Extend
Students add radiation rate or container material to make the model more comprehensive.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Has this ever happened to you? You make the perfect hot cocoa and then... it gets cold."*

Watch For

Bring in a real hot drink and a thermometer. Show the temperature dropping in real time.

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to find out WHERE the heat goes — and how to stop it."*

Watch For

Have students read objectives. Pre-teach 'conduction,' 'convection,' and 'insulation.'

Slide 3: Big Question

3 min

Say: *"Your cocoa starts at 85°C and ends at room temperature. That heat energy didn't disappear — so where did it go?"*

Watch For

Think-pair-share: Where do students think the heat goes? Capture predictions.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model to understand exactly how heat escapes — and then stop it."*

Watch For

Preview LEVER steps. Explain that the STEM challenge will use model evidence.

Slide 5: Activity 1: Components

8 min

Say: *"What factors affect how fast your cocoa cools? Some we can control, some we can't."*

Watch For

Guide sorting. Discuss why temperature difference and air movement are external inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"More insulation means LESS heat loss — that's a negative relationship. But more air movement means MORE heat loss."*

Watch For

Use hand gestures: positive = both go up together; negative = one up, other down.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare an open mug, a travel mug, and a mug in the wind!"*

Watch For

Students predict cooling curves first. Then run simulations and compare to predictions.

Slide 8: Discoveries

5 min

Say: *"So heat flows from hot to cold — ALWAYS. We can slow it down but never stop it completely."*

Watch For

Discuss the Second Law of Thermodynamics in kid-friendly terms.

Slide 9: STEM Challenge

12 min

Say: *"Now you're engineers. Design a container that keeps cocoa hot for 2 hours!"*

Watch For

Provide materials. Groups design, build, and test insulated containers.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Temperature Difference, Air Movement
- **INTERNAL:** Insulation Level, Container Surface Area

Relationships

- **Temperature Difference to Heat Loss Rate:** POSITIVE (+) — The bigger the gap between hot cocoa and room temperature, the faster heat flows out. As the drink cools and the gap shrinks, heat loss slows down naturally.
- **Insulation Level to Heat Loss Rate:** NEGATIVE (-) — Better insulation slows thermal energy transfer by trapping air and blocking conduction. A well-insulated container dramatically reduces cooling rate.
- **Air Movement to Heat Loss Rate:** POSITIVE (+) — Moving air carries away warm air molecules and replaces them with cool ones (convection). Wind or a fan dramatically increases the rate of heat loss.

Key Discoveries

- Heat ALWAYS flows from hot to cold — never the other way — until temperatures equalize
- Insulation doesn't add heat; it just slows down the transfer to the environment
- More surface area means faster heat loss because there's more area for energy to escape
- Moving air carries heat away much faster than still air (convection accelerates cooling)

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Cold flows into warm objects"	Cold is not a substance that flows. Only thermal energy (heat) flows — always from hot to cold. When your cocoa cools, heat is leaving the drink, not cold entering it. 'Feeling cold' means your body is losing heat to the environment.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Insulation creates heat"	Insulation doesn't generate any energy. It simply slows down the rate of heat transfer. A jacket keeps you warm by trapping your body's heat, not by creating warmth. If you put a coat on a snowman, it would actually melt SLOWER.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Metal objects are colder than wooden objects"	At room temperature, metal and wood are the SAME temperature. Metal just FEELS colder because it conducts heat away from your hand much faster than wood does. Your brain interprets rapid heat loss as 'cold.'